



Promoter Boundaries for the *luxCDABE* and *betIBA-proXWV* Operons in *Vibrio harveyi* Defined by the Method Rapid Arbitrary PCR Insertion Libraries (RAIL)

Christine M. Hustmyer,^a Chelsea A. Simpson,^a Stephen G. Olney,^a Douglas B. Rusch,^c Matthew L. Bochman,^b Julia C. van Kessel^a

^aDepartment of Biology, Indiana University, Bloomington, Indiana, USA

^bDepartment of Molecular and Cellular Biochemistry, Indiana University, Bloomington, Indiana, USA

^cCenter for Genomics and Bioinformatics, Indiana University, Bloomington, Indiana, USA

ABSTRACT Experimental studies of transcriptional regulation in bacteria require the ability to precisely measure changes in gene expression, often accomplished through the use of reporter genes. However, the boundaries of promoter sequences required for transcription are often unknown, thus complicating the construction of reporters and genetic analysis of transcriptional regulation. Here, we analyze reporter libraries to define the promoter boundaries of the *luxCDABE* bioluminescence operon and the *betIBA-proXWV* osmotic stress operon in *Vibrio harveyi*. We describe a new method called rapid arbitrary PCR insertion libraries (RAIL) that combines the power of arbitrary PCR and isothermal DNA assembly to rapidly clone promoter fragments of various lengths upstream of reporter genes to generate large libraries. To demonstrate the versatility and efficiency of RAIL, we analyzed the promoters driving expression of the *luxCDABE* and *betIBA-proXWV* operons and created libraries of DNA fragments from these loci fused to fluorescent reporters. Using flow cytometry sorting and deep sequencing, we identified the DNA regions necessary and sufficient for maximum gene expression for each promoter. These analyses uncovered previously unknown regulatory sequences and validated known transcription factor binding sites. We applied this high-throughput method to *gfp*, *mCherry*, and *lacZ* reporters and multiple promoters in *V. harveyi*. We anticipate that the RAIL method will be easily applicable to other model systems for genetic, molecular, and cell biological applications.

IMPORTANCE Gene reporter constructs have long been essential tools for studying gene regulation in bacteria, particularly following the recent advent of fluorescent gene reporters. We developed a new method that enables efficient construction of promoter fusions to reporter genes to study gene regulation. We demonstrate the versatility of this technique in the model bacterium *Vibrio harveyi* by constructing promoter libraries for three bacterial promoters using three reporter genes. These libraries can be used to determine the DNA sequences required for gene expression, revealing regulatory elements in promoters. This method is applicable to various model systems and reporter genes for assaying gene expression.

KEYWORDS reporter fusion, isothermal DNA assembly, arbitrary PCR, *Vibrio harveyi*, quorum sensing, bioluminescence, gene reporters

Central to the study of bacterial physiology and development is the ability to monitor and quantify gene expression. Monitoring of gene expression is greatly aided by the use of gene reporter fusions. Transcriptional and translational fusion constructs facilitate single-cell and population-wide gene expression investigations to

Received 30 November 2017 **Accepted** 1 March 2018

Accepted manuscript posted online 12 March 2018

Citation Hustmyer CM, Simpson CA, Olney SG, Rusch DB, Bochman ML, van Kessel JC. 2018. Promoter boundaries for the *luxCDABE* and *betIBA-proXWV* operons in *Vibrio harveyi* defined by the method rapid arbitrary PCR insertion libraries (RAIL). *J Bacteriol* 200:e00724-17. <https://doi.org/10.1128/JB.00724-17>.

Editor Victor J. DiRita, Michigan State University

Copyright © 2018 American Society for Microbiology. All Rights Reserved.

Address correspondence to Julia C. van Kessel, jcvk@indiana.edu.

For a commentary on this article, see <https://doi.org/10.1128/JB.00039-18>.

study the influence of regulatory factors, perform genetic screens, and visualize protein localization patterns. Typically, such reporters are cloned downstream of regulatory promoters or genes of interest and introduced into a model bacterial system, either on replicating plasmids or integrated into the genome. Numerous reporter genes have traditionally been used to assay the expression of genes such as *lux* (bacterial luciferase), *lacZ* (β -galactosidase), *phoA* (alkaline phosphatase), *bla* (β -lactamase), and *cat* (chloramphenicol acetyltransferase) (1, 2). However, the advent of more modern techniques has allowed for the use of fluorescent proteins such as green fluorescent protein (GFP) for these studies without the need for substrates or specialized medium (1–4).

To adequately and efficiently study the expression pattern of a particular gene, the defined regulatory region controlling promoter activity must be known. The region upstream of the promoter driving *luxCDABE* transcription in *Vibrio harveyi* is an example of a locus with a large and undefined regulatory region, which has limited studies of gene regulation. This particular promoter is of interest because it drives expression of the bioluminescence genes with a >100-fold increase in transcription and a >1,000-fold increase in bioluminescence production under activating conditions (i.e., quorum sensing) (5–10). It was previously suggested that the *lux* promoter requires ~400 bp upstream of the translation start site and ~60 bp downstream of the start codon for full activation of the *cat* reporter gene (9). The requirement for a large promoter region is due in part to the presence of seven binding sites for the transcription factor LuxR upstream of the primary transcription start site, each of which is necessary for maximal activation of the promoter (5, 6, 9). The ~400-bp region of $P_{luxCDABE}$ is relatively large compared to some bacterial regulatory promoters (e.g., the *lac* promoter) but comparable in size to other promoters with evidence of cooperative binding between transcription factors and DNA looping (e.g., the *araBAD* promoter) (5, 11–13). Indeed, full activation of the *luxCDABE* promoter requires the transcription factor LuxR and nucleoid-associated protein integration host factor (IHF), and DNA looping by IHF is proposed to drive interactions between LuxR and RNA polymerase (RNAP) for transcription activation (6).

Another *V. harveyi* operon that has an unknown promoter region is *betIBA-proXWV*. The *betIBA-proXWV* osmoregulation genes encode proteins required for the synthesis and transport of the osmoprotectant glycine betaine (14). These genes are autoregulated by the BetI repressor and activated 3- to 10-fold by LuxR (14). There are two sites in the *betIBA-proXWV* promoter that have been shown to be bound by LuxR *in vitro* and *in vivo*, though the role of these sites in transcriptional regulation has not yet been tested (5, 14). For both the *luxCDABE* and *betIBA-proXWV* operons, the boundaries of the promoters are not defined, and thus, mechanistic studies of transcriptional regulation of these operons is limited.

Here, we describe a new method for rapidly generating reporter plasmids that we used to define promoter regions. The rapid arbitrary PCR insertion libraries (RAIL) method exploits the power of arbitrary PCR and isothermal DNA assembly (IDA) to insert semirandomized fragments of promoter DNA into reporter plasmids (15–17). Using RAIL, we generated libraries containing fragments of various lengths of the region upstream of the *luxCDABE* operon transcriptionally fused to *gfp*. We used flow cytometry sorting to screen the library of promoter fragments for reporter expression and next-generation sequencing to map the 3' boundary of the *luxCDABE* promoter required for full activation. We also applied this method to two additional promoter regions in *V. harveyi* (*betIBA-proXWV* and *VIBHAR_06912*), and we demonstrated the versatility of the system by using two additional reporters, mCherry and β -galactosidase. This approach enabled us to identify the regions required for gene expression for multiple promoters and simultaneously produce usable gene reporter constructs. Our method should be widely applicable to any system for which gene reporters have been established and represents a simple and efficient technique to construct reporter fusions for molecular, genetic, and cell biology studies.

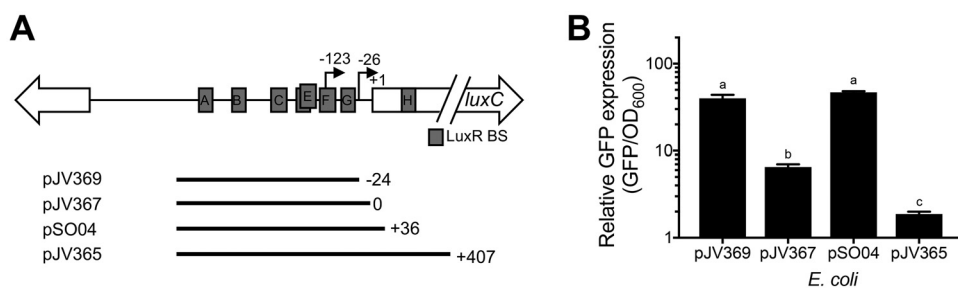


FIG 1 Promoter fusion plasmids for the *luxCDABE* genes. (A) Diagram of the regions of the *luxCDABE* promoter present in the plasmids listed. LuxR binding sites (BS) are shown as gray boxes, and the letters correspond to LuxR binding sites A through H. Transcription start sites are indicated by black arrows. The LuxC translation start site is shown as +1. Lengths of constructs are shown relative to the LuxC translation start site. (B) Relative GFP expression per unit of OD₆₀₀ (GFP/OD₆₀₀) for *E. coli* strains containing plasmids with various *luxCDABE* promoter fragments fused to *gfp* as indicated in panel A. The strains also contain either a plasmid constitutively expressing LuxR (pKM699) or an empty vector (pLAFR2). Relative expression was calculated by dividing the values for the pKM699-containing strain by the pLAFR2-containing strain. Different letters (a, b, and c) indicate significant differences ($P < 0.05$; one-way analysis of variance [ANOVA], followed by Tukey's multiple-comparison test of log-transformed data; $n = 3$).

RESULTS

Measurement of transcription activation from the *V. harveyi luxCDABE* promoter using fluorescent reporter fusions. To study the mechanism of LuxR regulation of the *luxCDABE* promoter, we constructed four reporter plasmids containing various fragments of the *luxCDABE* locus transcriptionally fused to *gfp* by traditional cloning methods (Fig. 1A). Each plasmid contains the same 5' end (~400 bp upstream of the *luxC* open reading frame [ORF]), and the 3' ends vary as follows: (i) 2 bp after the transcription start site at -26 (pJV369), (ii) at the LuxC translation start site (pJV367), (iii) 36 bp into the *luxC* ORF (pSO04), and (iv) 407 bp into the *luxC* ORF (pJV365) (Fig. 1A). All plasmid constructs contained the seven LuxR binding sites previously found to be essential for transcriptional activation. Only plasmid pJV365 contained LuxR site H, which has previously been shown to be nonessential for activation. We chose the lengths of these fragments to investigate the requirement for the 5' untranslated region (UTR), site H, and various lengths of the *luxC* ORF. We first tested LuxR activation of these reporter plasmids in *Escherichia coli* because expression of *luxR* in *E. coli* is sufficient to drive high levels of transcription of the *luxCDABE* operon (5, 10, 18), and the use of *E. coli* is more efficient for transformation. Activation of *luxCDABE* promoter transcription was assayed in *E. coli* strains containing either a second plasmid constitutively expressing *luxR* (pKM699) or an empty vector (pLAFR2). The plasmid containing the 3' boundary 36 bp into the *luxC* ORF was highly expressed (Fig. 1B), which is consistent with a previous study using a nearly identical promoter fragment (see Fig. S1A in the supplemental material) (5). The strain containing pJV369 with the DNA fragment up to and including the transcription start site also displayed high levels of GFP. Activation was appreciably decreased (~7-fold) for the pJV367 strain containing the 5' UTR but ending at the LuxC translation start site compared to the pSO04 strain (Fig. 1B). Also, the strain containing pJV365 with 407 bp of the *luxC* ORF was not activated >2-fold (Fig. 1B). A similar trend was obtained when these constructs were conjugated into *V. harveyi* strains and the GFP expression levels in the wild-type and $\Delta luxR$ mutant strains were compared (Fig. S1B). From these data, we conclude that a promoter fragment ending 2 bp past the transcription start site is sufficient for activation. Further, we revealed that various lengths of 3' constructs fused to *gfp* produce unexpected changes in gene expression across the *luxCDABE* promoter.

The RAIL method. Our observation that various 3' ends of the *luxCDABE* promoter greatly affected gene expression led us to expand our analysis of the expression profile of promoter fusions across the entire locus. Therefore, we needed to construct numerous promoter fragments transcriptionally fused to a fluorescent reporter. Instead of constructing each of these plasmids individually, we designed a cloning technique

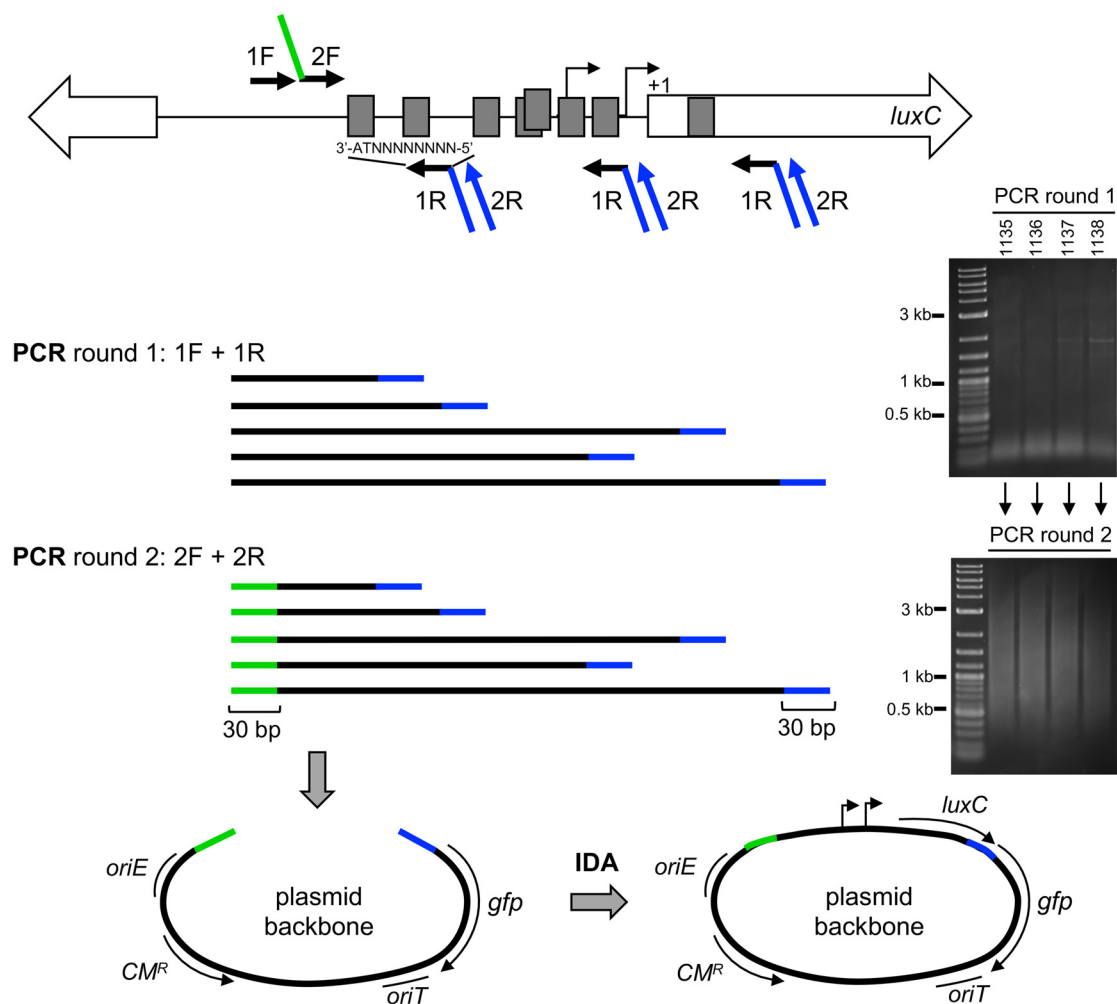


FIG 2 Schematic of the RAIL method for constructing libraries of promoter fusions. In PCR round 1, primers 1F and 1R were used to amplify a range of products specific to the *luxC* promoter. The 1R primers have eight random nucleotides (N) incorporated, are anchored by two nucleotides (AA, AT, TA, or TT), and contain a linker (shown in blue). In PCR round 2, the 2F and 2R primers were used to further amplify and add a linker to the products. Primer 2F anneals just downstream of 1F and contains a linker (shown in green). Primer 2R anneals to the linker region of primer 1R. The linear plasmid backbone was prepared by either restriction digestion or PCR. The library of products was inserted into the linear plasmid backbone via IDA with the homologous sequences present in the two linker regions (green and blue). The final plasmids contain fragments of the *luxC* promoter fused to the reporter. The gel images shown are examples of products from PCR rounds 1 and 2 for the *luxCDABE* locus with arbitrary R1 primers JCV1135-1138 and F1 primer SO71 in round 1 (Table S1). The products of the round 1 reactions were used in round 2 with primers SO72 and JCV1139 (Table S1).

combining the power of arbitrary PCR and IDA (also called Gibson assembly) (15–17). This method enabled us to simultaneously amplify fragments of various lengths and clone them into a vector backbone to create a library in four simple steps (Fig. 2). First, arbitrary primers were used in a preliminary round of PCR in conjunction with a primer that specifically anneals to the promoter (Fig. 2, primer 1F). Four arbitrary primers were synthesized with eight sequential random nucleotides anchored at the 3' end with two specific nucleotides: AA, TT, AT, or TA (Table S1). We chose to use A-T pairs to anchor the primer because of the low G+C content of *V. harveyi*. Each of these four primers also contains a linker at the 5' end (Fig. 2, primer 1R). The first round of PCR produced a range of products that varied in length from 100 to >3,000 bp and appeared as faint smears of products, as expected for random priming (Fig. 2). For some loci, no smear could be visualized by gel electrophoresis after the first round of PCR, but this did not impact the success of the second round of amplification.

In the second step, the products from round 1 were further amplified with a nested primer (primer 2F) and a linker with homology to the plasmid backbone was added

(Fig. 2). Primer 2R anneals to the linker on primer 1R. The second round of PCR using these primers was performed with the products from round 1 as templates. This second step served to increase the amount of DNA product and to add a linker to the 5' end. Each reaction in round 2 produced a smear of products that contained homology to the plasmid backbone at their 5' and 3' ends (Fig. 2). The smear of products can also be gel extracted to the desired size. In the third step, the plasmid backbone was PCR amplified or digested by specific restriction enzymes to form a linear product (Fig. 2). In the fourth and final step, IDA was performed to clone the promoter fragments into the plasmid backbone and the mixture was transformed into *E. coli* to obtain isolated clones (Fig. 2).

Defining the 3' boundary of the *luxCDABE* operon by RAIL. We used the RAIL method to generate a large library of plasmids with promoter fragments fused to *gfp*. This library had fixed 5' ends and various 3' ends generated by combining PCR products from four arbitrary primers, as shown in Fig. 2, and inserts ranging from ~50 to >1,000 bp. We screened for *gfp* activation by fluorescence-activated cell sorting (FACS). The libraries were sorted by FACS into four groups with no, low, medium, and high GFP expression (Fig. 3A). The no-GFP pool contained cells expressing fluorescence similar to or lower than that of the negative-control strain. The high-GFP pool contained cells with fluorescence expression similar to or higher than that of the positive-control strain. The low- and medium-GFP pools were arbitrarily chosen to collect cells in the intermediate region between no GFP and high GFP without any overlap among the four bins (Fig. 3A). Illumina sequencing of the plasmid DNA from these pools enabled us to visualize the 3'-terminal end of the region cloned into the plasmid by graphing the location of the sequencing coverage (42 bp) and the 3'-terminal nucleotides (Fig. 3B; Fig. S2A). From these graphs, we pinpointed the boundary in the *luxCDABE* promoter required for maximum expression and showed the expression profile for promoter fragments across the entire locus (Fig. 3B; Fig. S2A). The plasmids containing promoter fragments that terminated at nucleotide +129 (relative to +1, the start of the *luxC* ORF) were highly enriched in the high-expression pool, and plasmids in the no-expression pool were specifically de-enriched in this same location (Fig. 3B; Fig. S2A). The high-expression pool had a clear 3' boundary at +129, which is 16 bp upstream of LuxR site H. Thus, a DNA fragment that terminates at +129 includes LuxR sites A, B, C, D, E, F, and G (6). The observation that LuxR site H was not included in this region of high expression is consistent with previous findings that site H is nonessential for transcription activation at high cell density in *V. harveyi* (6). We conclude from these data that fragments with 3' ends longer than +129 were decreased in reporter gene expression. There is also a clear edge where sequencing coverage drops off for the high-expression pool at -55 (Fig. 3B). However, the exact minimum boundary cannot be determined because we did not use every combination of anchor nucleotides in the arbitrary primers. Also, within the high-expression pool, we noted a peak of sequencing coverage that started at +36, suggesting that this is the minimum length promoter sufficient for high GFP expression in this library (Fig. S2A). Thus, promoter fragments with ends ranging from +36 through +129 yield maximum expression levels without being detrimental. The medium-expression pool contained sequences that terminated at +199, which is located 32 bp beyond LuxR site H (Fig. 3B). Plasmids with promoter fragments that extended throughout the *luxC* ORF past +199 had low levels of expression, whereas plasmids without GFP expression were limited to promoter regions upstream of -55 (Fig. 3B). We conclude that long promoter fragments decrease GFP expression and are not suitable reporter plasmids. We also conclude that plasmids containing promoter fragments shorter than -55 are not sufficient to activate transcription. Collectively, by analyzing the sequencing data of RAIL libraries, we located the DNA region that is sufficient for maximal transcription activation (-393 to +36), validated previous findings that LuxR sites A through G are required for activation of *luxCDABE* (6), and defined the expression profile for the *luxCDABE* locus in the context of a transcriptional fusion to *gfp*.

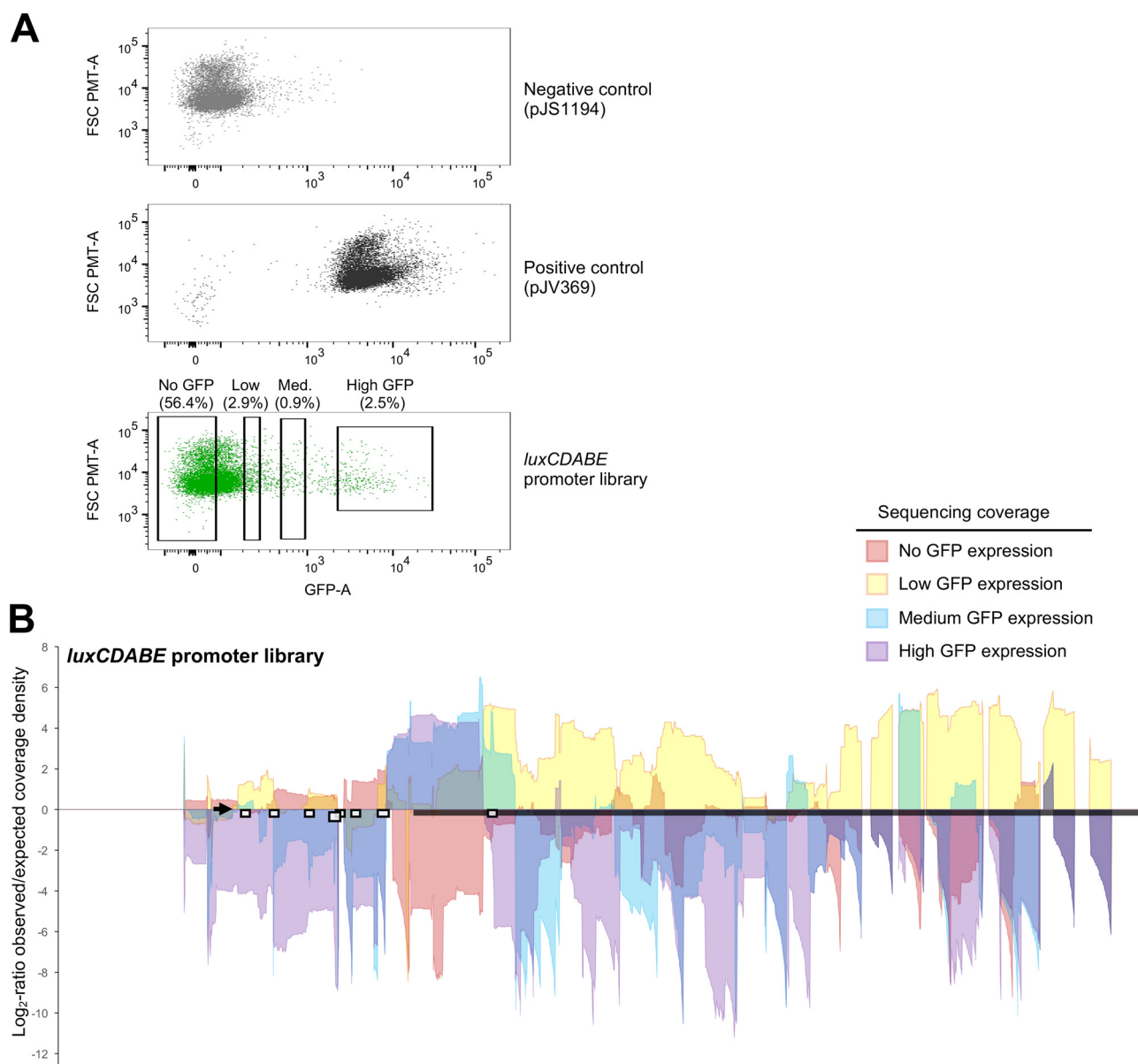


FIG 3 Flow cytometry sorting and next-generation sequencing of the $P_{luxCDABE}$ -*gfp* library. (A) FACS GFP expression data from three *E. coli* cultures, (i) a negative-control strain containing an empty-vector control (pJS1194) and a plasmid expressing LuxR (pKM699), (ii) a positive-control strain containing a $P_{luxCDABE}$ -*gfp* reporter plasmid (pJV369) and pKM699, and (iii) the $P_{luxCDABE}$ library of plasmids in *E. coli* containing pKM699. Gates (boxes) indicate the cells sorted (percentage of the total population) into four bins with no, low, medium, and high GFP expression. Data were obtained with FlowJo software. (B) Genomic regions associated with differential expression of the *luxCDABE* locus. The nucleotide coverage of the reads (42 bp) is shown for different populations of cells with distinct levels of GFP reporter expression as indicated in the color key. Data are graphed as the nucleotide position (x axis) versus the log₂ ratio of observed coverage density divided by the expected coverage density (as determined by the read counts observed in the total library). Areas with positive values in log₂ observed/expected coverage densities indicate an enrichment of sequence reads in that region, and areas with negative values indicate lower-than-expected-frequency reads. The thick black bar indicates the location of the *luxC* ORF. The locations of LuxR binding sites are indicated by white boxes. The locus to which the 2F primer anneals is indicated by the arrow.

Defining the 3' boundary of the *betIBA-proXWV* operon by RAIL. We next used the RAIL strategy to construct reporter clones for the *betIBA-proXWV* operon with a different fluorescent reporter, *mCherry*. We screened the promoter clones individually before using the high-throughput flow cytometry method to analyze the library to test whether we could identify useful promoter clones via small-scale screening. Approximately 40 plasmids were screened by restriction digestion for inserts of various sizes, and the inserts were sequenced to determine the size of the inserted region. We observed that plasmids containing regions shorter than the predicted transcription

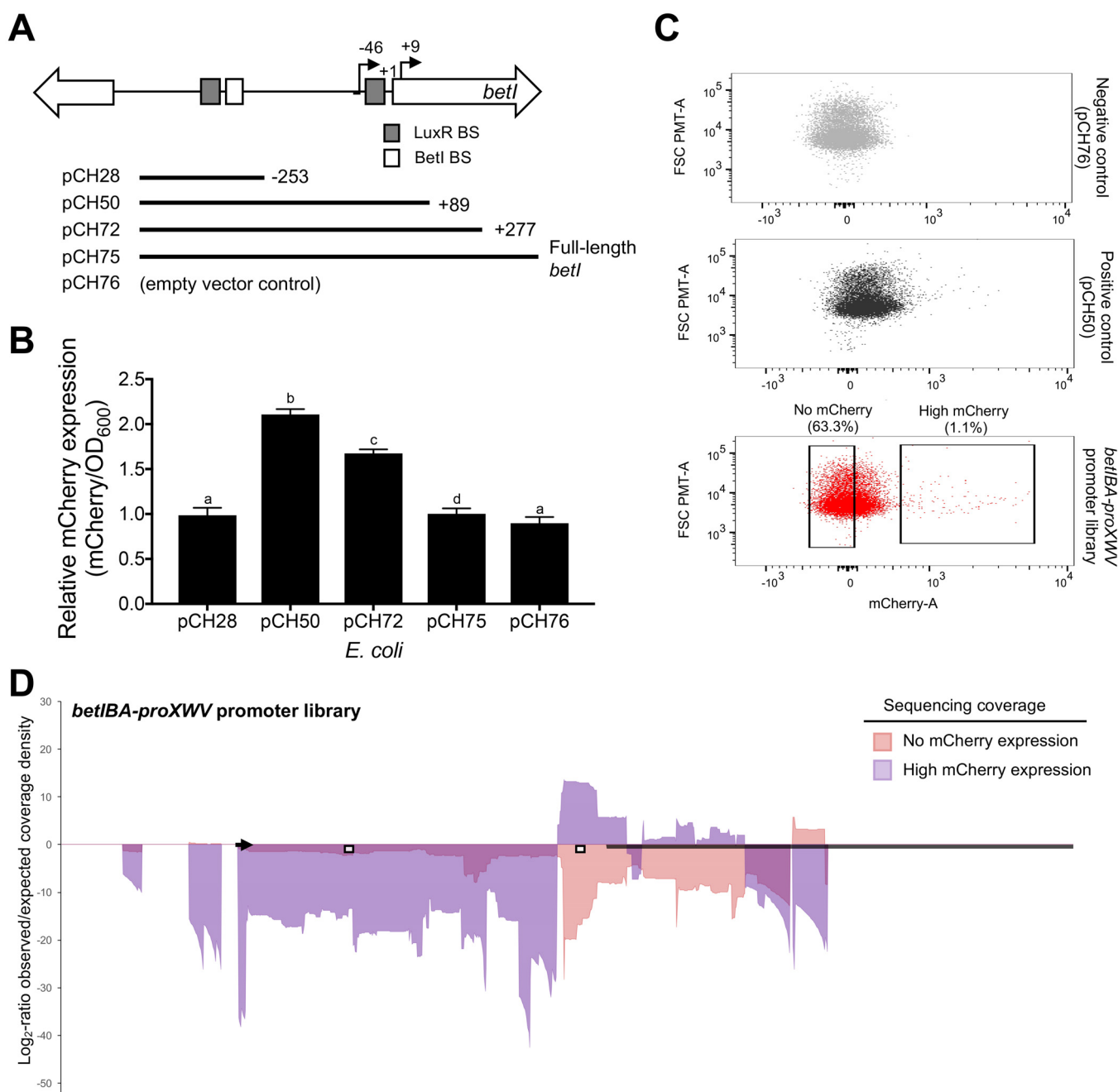


FIG 4 Expression data from the *P_{betIBA-proXWV}-mCherry* library. (A) Diagram of the regions of the *betI*-*proXWV* promoter present in various plasmids. LuxR binding sites (BS) and the BetI binding site are indicated. Putative transcription start sites are indicated by black arrows. The BetI translation start site is shown as +1. Lengths of constructs are shown relative to the BetI translation start site. (B) Relative *mCherry* expression per unit of OD₆₀₀ (mCherry/OD₆₀₀) for *E. coli* strains as calculated by dividing the values for the pKM699-carrying strain by those for the pLAFR2-carrying strain. The strains contained plasmids with various *betI*-*proXWV* promoter fragments fused to *mCherry*, as indicated in panel A. Different letters (a, b, c, and d) indicate significant differences ($P < 0.05$; one-way ANOVA, followed by Dunnett's multiple-comparison test; $n = 3$). (C) FACS data showing *mCherry* expression for three *E. coli* cultures, (i) a negative-control strain containing pCH76 and pKM699, (ii) a positive-control strain containing pCH50 and pKM699, and (iii) the *P_{betIBA-proXWV}* library in *E. coli* containing pKM699. Gates indicate the cells sorted (percentage of the total population) into two bins with no and high *mCherry* expression. (D) Genomic regions associated with differential expression of the *betI*-*proXWV* locus. The nucleotide coverage of the reads (42 bp) is shown for different populations of cells with distinct levels of *mCherry* reporter expression as indicated in the color key. Data are graphed as the log₂ ratio of observed coverage density divided by the expected coverage density (as determined by the read counts observed in the total library). Areas with positive values in log₂ observed/expected coverage densities indicate an enrichment of sequence reads in that region, and areas with negative values indicate lower-than-expected-frequency reads. The thick black bar indicates the location of the *betI* ORF. The locations of LuxR binding sites are indicated by white boxes. The locus to which the 2F primer anneals is indicated by the arrow.

start sites did not show any activation compared to the empty-vector control strain (Fig. 4B, pCH28 as an example). However, plasmids with larger regions that extended into the *betI* ORF were activated by LuxR, such as pCH50 and pCH72 (Fig. 4B). Plasmids containing the entire *betI* gene did not display activation (Fig. 4B, pCH75). These data show that the RAIL method can be used for small-scale screening for promoter clones by individually assaying plasmids.

We next synthesized a large library of *betIBA-proXWV* promoter fusions to *mCherry* by using RAIL. It is important to note that only one arbitrary primer was used to generate this library, which limited the range of PCR products across the locus. This library of clones was sorted by FACS for those that maximally expressed *mCherry* (Fig. 4C). The dynamic range of the *betIBA-proXWV* promoter is substantially smaller than that of *luxCDABE*, resulting in an approximately 3-fold difference in expression in the averages of the positive and negative controls. Thus, we chose to sort cells with fluorescence levels above that of the negative-control strain (Fig. 4C). In doing so, we lost cells that exhibited intermediate fluorescence levels but could therefore be assured that all of the cells we collected were expressing high levels of fluorescence. The Illumina sequencing coverage and 3'-terminal nucleotides of the DNA in the two pools was graphed (Fig. 4D; Fig. S2B). Sequencing analyses revealed the minimum 3' boundary of the *betIBA-proXWV* promoter to be at -13 (Fig. 4D; Fig. S2B) relative to $+1$, the start of the *betI* ORF, suggesting that the -46 transcription start site is the primary site for this locus. The high-expression pool contained plasmids with DNA fragments up through the first portion of the *betI* ORF at $+25$, which then tapered off (Fig. 4D; Fig. S2B). Plasmids with fragments that extended more than halfway through the *betI* gene displayed low or no expression. Collectively, these data showed that similar to the *luxCDABE* locus, transcription reporters were functional if they contained DNA fragments past the 3' boundary near the transcription start site. However, longer fragments extending into the ORF decreased reporter gene expression.

Versatility of the RAIL method for cloning with other promoters and reporter genes. We also successfully used the RAIL technique to generate a promoter library by using the *lacZ* reporter for another *V. harveyi* gene, *VIBHAR_06912*, which encodes a transcription factor. *VIBHAR_06912* expression is repressed by LuxR (19), and this is likely indirect repression because there are no detectable LuxR binding sites in this region (5). Using RAIL, multiple clones with various promoter lengths were generated as transcriptional fusions to *lacZ*, and *V. harveyi* strains were assayed for β -galactosidase activity (Fig. 5A). All of the plasmids with long promoters were repressed by LuxR in the wild-type strain compared to the $\Delta luxR$ mutant strain (Fig. 5B). Conversely, a plasmid with a short fragment (pJV342) showed the same level of β -galactosidase activity in the wild-type strain as in the $\Delta luxR$ mutant strain (Fig. 5B). We note that each construct promoted transcription to a different level, even in the absence of LuxR repression. This suggests that other regulatory elements in addition to LuxR affect transcription at this locus. Thus, we conclude that we again generated functional promoter fusion plasmids for this promoter for future studies of gene expression and regulation of *VIBHAR_06912*.

Reporter gene affects measurement of gene expression. We noted that for each of the three promoters we studied, plasmid constructs that contained promoter regions that extended into the ORF of the first gene had variable levels of expression. For example, the pJV365 plasmid that included 407 bp of the *luxC* gene only expressed GFP ~ 2 -fold more in the presence of LuxR than in its absence (Fig. 1B). This is in contrast to plasmid pMGM115 from the study of Miyamoto et al., which contains the full *luxC* ORF and displays maximal activation of the *cat* gene (Fig. S1A) (~ 50 -fold more than truncated promoters) (9). To examine these contradictory results further, we constructed plasmids containing the entire *luxC* gene and its promoter region driving *gfp*, *lacZ*, or *mCherry* expression (Fig. 6A and B). These constructs contained the intragenic region between *luxC* and *luxD* (15 bp), and the reporter gene was cloned in place of the *luxD* ORF (Fig. 6B). We observed that the *lacZ* and *mCherry* plasmids were activated 16- to 20-fold, whereas the *gfp* construct was activated only 1.6-fold by LuxR in *E. coli*

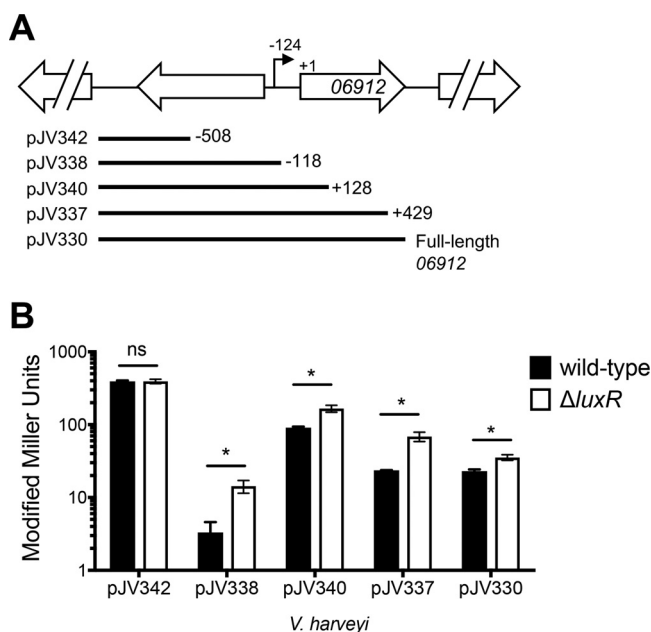


FIG 5 *VIBHAR_06912* promoter-reporter fusion plasmids. (A) Diagram of the regions of the *VIBHAR_06912* promoter present in various plasmids. The putative transcription start site is indicated by a black arrow. The *VIBHAR_06912* translation start site is shown as +1. Lengths of constructs are shown relative to the *VIBHAR_06912* translation start site. (B) Modified Miller units are shown for the wild-type (BB120) and $\Delta luxR$ mutant (KM669) strains containing various plasmids as indicated in panel A. Asterisks indicate significant differences between the wild-type and $\Delta luxR$ mutant strains with the various plasmids ($P < 0.05$; ns, not significant; $n = 3$; two-way ANOVA, followed by Sidak's multiple-comparison test of log-transformed data).

(Fig. 6C). The *gfp* (pSO05) and *mCherry* (pSO11) plasmids had similar levels of activation when they were introduced into wild-type *V. harveyi*, though neither was expressed maximally (Fig. S1B and C).

We hypothesized that the observed decrease in activation with longer fragments might be due to instability of the transcript when the *luxC* ORF is present upstream of the *gfp* reporter. Thus, we constructed *mCherry* reporter plasmids containing the same four *luxCDABE* promoter fragments that were fused to *gfp* in Fig. 1A and assayed these in *E. coli* (Fig. 6A). We verified that the shortest region tested (2 bp past the primary transcription start site) was sufficient for activation, and there was no significant difference in expression with a construct containing a slightly longer promoter fragment (Fig. 6D), and the three shortest fragments were activated >50-fold (Fig. 6D). However, as seen with GFP, the plasmid with the longest promoter fragment (e.g., pJV366 with 407 bp of the *luxC* ORF) yielded a significantly lower level of *mCherry* expression (Fig. 6D) (activated 17-fold), which was similar to the construct containing the entire *luxC* ORF (Fig. 6C, pSO11) (activated 17-fold). A similar trend was observed in *V. harveyi* with these *mCherry* plasmids (Fig. S1C). Thus, we conclude that constructs containing long fragments indeed decrease the expression of downstream reporters, and with some of these, large decreases occur (i.e., *gfp*). This result is not observed with expression of the *luxCDABE* operon *in vivo*; the expression levels of the five genes in the operon are similar and do not differ by more than 2-fold from one another, as determined by microarray analysis (19).

To examine these results, we measured *gfp* transcript levels for several P_{luxC} reporter plasmids in *E. coli*. The relative *gfp* transcript levels from quantitative reverse transcription (qRT)-PCR measurements were high for the three plasmids containing short regions of the *luxCDABE* promoter, but as seen with GFP expression measurements, *gfp* transcript levels were significantly ~43-fold lower in a strain containing the pJV365 plasmid containing 407 bp into the *luxC* ORF than in a strain containing pJV369 (Fig. 6E). The *gfp* transcript levels were significantly lower in pJV365 than in all of the other

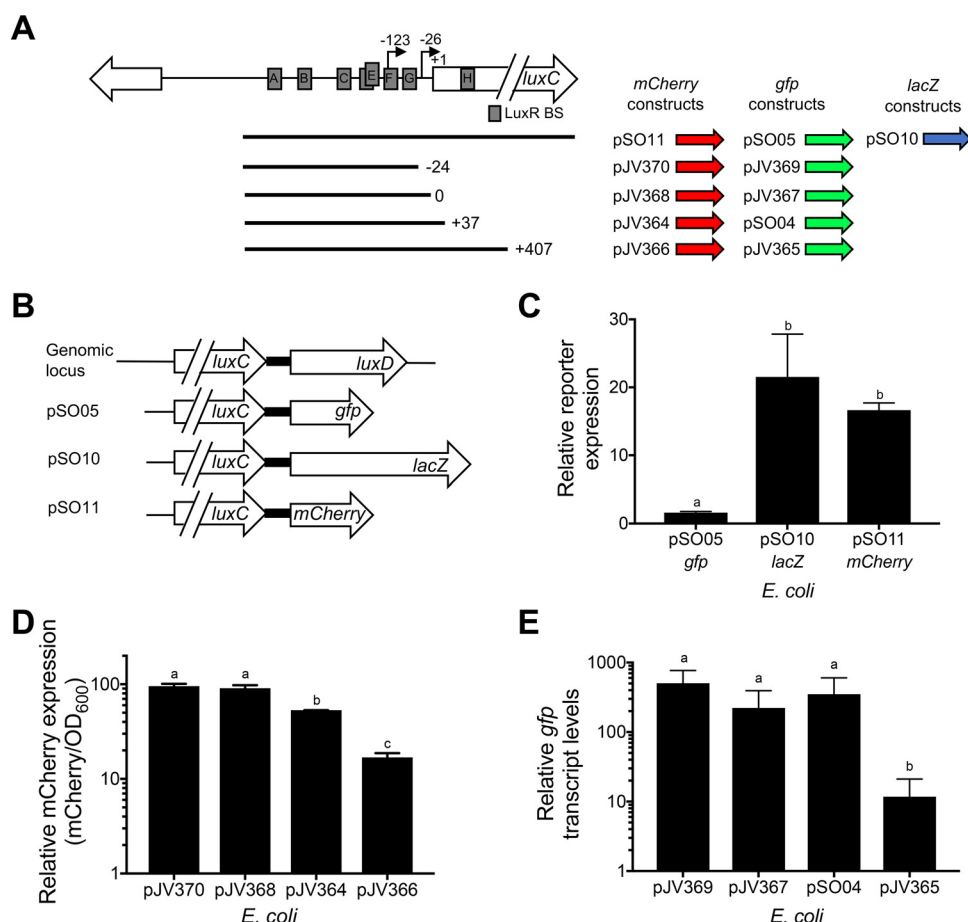


FIG 6 Measuring transcription of plasmids with long promoter regions fused to reporters. (A) Diagram of the regions of the *luxCDABE* promoter present in various constructs. LuxR binding sites (BS) are indicated. Transcription start sites are indicated by black arrows. The LuxC translation start site is shown as +1. Lengths of constructs are shown relative to the LuxC translation start site. Plasmids containing *mCherry*, *gfp*, or *lacZ* fusions are indicated. (B) Diagram of plasmids containing *luxCDABE* promoters and the *luxC* ORF fused to *gfp* (pSO05), *lacZ* (pSO10), or *mCherry* (pSO11). Each construct contains the 15-bp sequence between *luxC* and *luxD* as shown. (C) Relative expression of reporters (*gfp*, *mCherry*, and *lacZ*) is shown for *E. coli* strains containing either a plasmid constitutively expressing LuxR (pKM699) or an empty vector (pLAFR2). Relative expression was calculated by dividing the values for the pKM699-containing strain by the pLAFR2-containing strain. LacZ expression was determined by modified Miller assays, and *gfp* and *mCherry* expression was assayed with a plate reader. Different letters (a, b) indicate significant differences ($P < 0.05$; one-way ANOVA, followed by Tukey's multiple-comparison test of log-transformed data; $n = 3$). (D) Relative expression of *mCherry* (mCherry/OD₆₀₀) in *E. coli* strains containing either pKM699 or pLAFR2 calculated as described for panel C. Different letters (a, b, and c) indicate significant differences ($P < 0.05$; one-way ANOVA, followed by Tukey's multiple-comparison test of log-transformed data; $n = 3$). (E) Relative *gfp* transcript levels determined by qRT-PCR in *E. coli* strains containing either pKM699 or pLAFR2 calculated as described for panel C. Different letters (a and b) indicate significant differences ($P < 0.05$; one-way ANOVA, followed by Tukey's multiple-comparison test of log-transformed data; $n = 6$).

plasmids with shorter promoter fragments tested. Thus, we conclude that the decrease in GFP expression in the pJV365-containing strain is due to a decrease in transcript levels, which may be caused by either transcript instability or a decrease in transcription initiation or elongation in plasmids with long promoter fragments. We did not observe a significant decrease in *gfp* transcript levels with pJV367 as observed with GFP expression (Fig. 1B), suggesting that the decrease in GFP expression may be due to constraints at the posttranscriptional or translational level. These results indicate that testing of multiple promoter fusions is beneficial for identifying a promoter-reporter fusion that functions *in vivo* to mimic expression from the native locus.

DISCUSSION

We have developed the RAIL method for rapid construction of promoter fusion plasmids and demonstrated that this approach can be applied to multiple promoters

and reporter genes. The RAIL strategy can be used to quickly generate a few reporters or to create large libraries of promoter fusions for high-throughput analysis of the regions that drive transcription activation. The method requires simple cloning steps, and once the system is designed for a particular plasmid backbone, only two locus-specific primers are needed. For our plasmid backbone, we designed arbitrary primer sets for creating fusions to *gfp*, *mCherry*, and *lacZ* that can be used with any gene locus (Table S1), and these primers can be easily modified for use in any plasmid with a reporter gene.

Our library sets revealed several important findings with regard to the expression profiles of the *luxCDABE* and *betIBA-proXWV* promoters. First, we validated previous work describing the requirement for LuxR binding sites in these promoters (5, 6, 9, 14). Second, we identified the promoter region that is required for high levels of transcription activation for these two promoters. We did not resolve the 3' boundary to a specific nucleotide locus in these experiments because we did not use every combination of anchor nucleotides in the arbitrary primers and restricted our analysis to combinations of A and T pairs. To acquire complete coverage, a full set of random primers with every combination of nucleotides as anchors should be used. However, with this resolution, we clearly found a marked difference in plasmids containing various fragments of the promoters such that we could identify the region sufficient for maximum gene expression. Even when only one arbitrary primer was used, as in the case of the *betI* library, we were still able to determine regions of maximal regulation, but with less coverage. Therefore, we have shown that the RAIL method is applicable for small-scale studies in which perhaps only one primer is used for quicker analysis or large-scale studies where a combination of arbitrary primers will result in greater coverage of the promoter region and produce higher resolution. Further, arbitrary primer design in these studies was limited to terminal A-T combinations because of low G+C content in *V. harveyi* (~45%). In GC-rich organisms, we propose that arbitrary primers should instead be designed with terminal G-C combinations for more precise anchoring. Smaller fragments may be sufficient to drive the same level of gene expression, which can be tested with the full series of anchor nucleotides in the arbitrary primers. Further, the minimum 5' ends of the promoters in this study are not known, so 5' ends were chosen several hundred base pairs upstream of the ORF (~400 to 1,000 bp). Future studies could use the same approach to map the 5' boundary of these two promoters, which is a separate but intriguing question.

Third, our data conclusively demonstrate that there is no requirement for the region downstream of the transcription start site for full activation of the *luxCDABE* promoter. This finding is important because a previous study by Miyamoto et al. also tested promoter regions for *luxCDABE* via a *cat* promoter (9). Among the various constructs tested in that study, the pMGM127 plasmid contains a region truncated slightly upstream of the -26 transcription start site (the specific 3' end is undefined in the article) and the pMGM116 plasmid includes a 3' end at +61 relative to the *luxC* start codon (Fig. S1A). The shorter promoter in pMGM127 shows no transcription activation, whereas the longer promoter in pMGM116 had full activation of the *cat* reporter (9). These data and other observations have led to an anecdotal hypothesis in the field that there is an element downstream of the transcription start site that is required for full activation of the *luxCDABE* promoter. Our data refute this hypothesis because the pJV369 plasmid does not include the 5' UTR and is maximally activated in both *E. coli* and *V. harveyi*.

Finally, our analysis of various promoter-reporter fusion plasmids demonstrated that not all reporter fusions are created equal and suggests that testing various reporter constructs for each gene of interest is beneficial to finding the optimal reporter for downstream assays. We noted that plasmid constructs with long fragments of the *luxCDABE* and *betIBA-proXWV* promoters that included sections of the first ORF in the operon were substantially decreased in expression, and we showed that this is effective at the transcript level for *luxC-gfp* fusions (Fig. 6E). However, we also noted that the strains containing the pJV367 plasmid that had a decrease in GFP fluorescence

did not exhibit a decrease in *gfp* transcript levels (Fig. 6E). This result implies that the 7-fold decrease in GFP fluorescence is due to posttranscriptional or translational effects such as mRNA secondary structure that may block translation initiation.

We have focused attention on our results for the multiple plasmids with long *luxCDABE* promoter fragments that show significantly decreased levels of reporter expression (pJV365, pJV366, pSO05, pSO10, and pSO11). Our qRT-PCR analysis showed that transcript levels were significantly lower for pJV365 than for its counterpart plasmids with shorter promoter fragments. These data suggest that decreased reporter expression for all of the other long-promoter plasmids may also be due to decreased transcript levels. There are at least two possible reasons why the pJV365 plasmid has decreased transcript levels. One possibility is that transcripts generated with fragments of the *luxCDABE* operon fused to the *gfp* gene fold into unstable secondary structure and be subject to degradation. However, we suspect that this explanation is unlikely to be the cause of low expression for every plasmid with a fragment longer than +129, as we would predict that at least some would be stable. A second possibility is that LuxR binding to site H is acting as a roadblock to transcription elongation, which results in the abrupt drop in GFP expression for plasmids containing promoter regions that terminate after site H. Previously, we showed that scrambling of site H does not decrease LuxR activation of β -galactosidase expression in a *luxC-lacZ* reporter plasmid under conditions in which LuxR is maximally expressed at high cell densities in *V. harveyi* (6). However, the results of our expression profiling experiment with *E. coli* with the *luxCDABE* promoter library suggest that plasmids that contain LuxR site H have decreased levels of transcription activation and are strictly in the low GFP expression pool (Fig. 3B). LuxR has an extremely high affinity for site H, with a K_d of 0.6 nM, one of the tightest LuxR binding affinities in the genome (5). Thus, it is curious that LuxR binds at this locus with no apparent activation defect when tested at high cell density in *V. harveyi*.

Protein roadblocks have been described in bacteria and eukaryotes that hinder transcription, and elongation factors aid in transcription elongation through these roadblocks by various mechanisms (e.g., Mfd in *E. coli*) (20). In addition, when multiple RNAP molecules are initiated from the same promoter, these trailing RNAP complexes can “push” a stalled RNAP through a roadblock (21). Thus, it is possible that higher levels of transcription initiation of the *luxCDABE* promoter in *V. harveyi* at high cell densities drive transcription elongation through site H, whereas lower levels of LuxR at low cell densities in *V. harveyi* or in our synthetic *E. coli* system are not sufficient to push through the LuxR site H roadblock. LuxR concentrations are low in the cell at low cell densities, and thus, the relatively few LuxR molecules likely bind to the highest-affinity sites, such as site H in the *luxC* ORF. As cells grow to high densities, LuxR accumulates (19, 22, 23) and enables LuxR binding to other sites, which drives high levels of transcription initiation and may relieve binding of LuxR to site H to allow RNAP elongation. Alternatively, the roadblock might be relieved by restructuring of the DNA architecture at the locus. Because we have already shown that IHF binds to multiple places in the *luxCDABE* region and its binding is positively cooperative with LuxR, this DNA bending may play a role in the removal of transcription roadblocks. We also observed a sharp difference between the medium- and low-expression pools just downstream of the LuxR site H (Fig. 3B), suggesting that there may be yet another roadblock in this region. Future studies should elucidate the role of LuxR binding sites within ORFs in *V. harveyi*, which are observed throughout the genome (5).

In conclusion, the RAIL method offers a rapid and efficient method to obtain libraries of reporter fusions that can be used for various studies of gene expression and regulation. Often in bacterial genetics, researchers attempt to create promoter fusions by cloning a reporter gene in place of the translation start site, and this would have yielded suboptimal reporters for the *luxCDABE* promoter. Anecdotally, and as we experienced with the *betIBA-proXWV* and *luxCDABE* promoters, one often needs to construct multiple reporter fusions to identify a promoter region that drives gene expression mimicking native locus gene expression. Thus, our method is more efficient

by generating numerous clones in a single cloning experiment. We envision the use of the RAIL method for numerous other purposes, such as creating functional GFP fusions for studying protein localization, identifying *cis*-regulatory sequences in promoters (e.g., protein binding sequences), inserting affinity tags for purification strategies, and identifying highly expressed soluble constructs for protein purification. Finally, this method should be applicable to any model organism for which genetic cloning techniques have been established.

MATERIALS AND METHODS

Bacterial strains and media. *E. coli* strains S17-1 λ pir, DH10B, and derivatives (see Table S2 in the supplemental material) were used for cloning and *in vivo* assays. *E. coli* strains were grown with shaking at 275 rpm at 30°C in lysogeny broth (LB) augmented with 10 μ g/ml chloramphenicol and 10 μ g/ml tetracycline when required. The *V. harveyi* BB120 is strain ATCC BAA-1116, which was recently reassigned to *Vibrio campbellii* (24). It is referred to as *V. harveyi* here for consistency with previous literature. BB120 and derivatives (Table S2) were grown at 30°C with shaking at 275 rpm in LB marine (LM) medium supplemented with 10 μ g/ml chloramphenicol when required. LM medium is prepared similarly to LB (10 g of tryptone, 5 g of yeast extract) but with 20 g of NaCl instead of the 10 g used for LB.

Molecular methods. Oligonucleotides (Table S1) were purchased from Integrated DNA Technologies. All PCRs were performed with Phusion HF polymerase (New England BioLabs) or iProof polymerase (Bio-Rad). Restriction enzymes, enzymes for IDA (15), and deoxynucleoside triphosphates (dNTPs) were obtained from New England BioLabs. DNA samples were visualized on 1% agarose gels. Standard cloning methods and primers for the single plasmid constructs listed in Table S3 are available upon request. Standard sequencing of single plasmid constructs was conducted by ACGT, Inc., and Eurofins Genomics. To measure the expression levels of fluorophore reporter plasmids, *E. coli* and *V. harveyi* strains were grown overnight at 30°C with shaking at 275 rpm. Strains were diluted 100-fold in growth medium and selective antibiotics in 96-well plates (black with a clear bottom), covered with microporous sealing tape (USA Scientific), and incubated at 30°C with shaking at 275 rpm for 16 to 18 h. Fluorescence and optical density at 600 nm (OD₆₀₀) of strains expressing *mCherry* and *gfp* were measured with either a BioTek Synergy H1 or a Cytation plate reader. Miller assays were conducted as previously described (6). RNA extraction and qRT-PCR were performed and analyzed as described previously (14) with primers listed in Table S1 on a StepOne Plus real-time PCR machine (Applied Biosystems). Transcript levels were normalized to the level of expression of the internal standard *recA*, and the standard-curve method was used for data analysis. The error bars on graphs represent the standard deviations of measurements of at least three biological samples. Statistical analysis was performed with GraphPad Prism version 7.0c. Additional information about statistical analyses pertinent to each result set is included in the figure legends.

RAIL: construction of promoter libraries by arbitrary PCR. The arbitrary PCR method was adapted from a study by Schmidt et al. (25), with several modifications. The first round of PCR was conducted with two primers, a forward primer specific to the promoter of interest and a reverse primer for random DNA amplification (Fig. 2, primers 1F and 1R, respectively). The 1R primer includes a priming sequence followed by eight random nucleotides ('N') and terminating in two defined nucleotide anchors, AT, TA, TT, or AA (Table S1). For PCR round 1, ~10 to 100 ng/ μ l genomic DNA from *V. harveyi* BB120 or a plasmid containing the region of interest was added as the template. The reaction mixture included 200 μ M dNTPs, 250 μ M primers, 5% dimethyl sulfoxide (DMSO), 0.5 μ l of Phusion polymerase, and 1 $\times$ Phusion buffer. Cycling parameters were as follows and as previously published (25): an initial denaturation at 95°C for 5 min; 5 cycles of 95°C for 30 s, 25°C for 30 s, and 72°C for 2.5 min; 30 cycles of 95°C for 30 s, 50°C for 30 s, and 72°C for 2.5 min; and a final extension step of 72°C for 10 min. PCRs were purified with the GeneJet PCR purification kit (Thermo Scientific) and eluted in 30 to 50 μ l of elution buffer. The second round of PCR used primers 2F and 2R (Fig. 2). The forward primer (2F) included 30 nucleotides of homology to the plasmid backbone for IDA and a sequence specific to the promoter of interest that is nested downstream of the 1F primer. The reverse primer (2R) included the priming sequence that is identical to that of primer 1R. To perform PCR round 2, 5 μ l of the purified DNA from round 1 was used as the template. These reaction mixtures also included 200 μ M dNTPs, 250 μ M primers, 5% DMSO, 0.5 μ l of Phusion polymerase, and 1 $\times$ Phusion buffer. The cycling parameters were as follows: an initial denaturation at 95°C for 5 min; 35 cycles of 95°C for 30 s, 50°C for 30 s, and 72°C for 2.5 min; and a final extension step of 72°C for 10 min. PCR products were separated by agarose gel electrophoresis and visualized by UV transillumination, and the products were gel extracted to the desired target size with a GeneJet gel extraction kit (Thermo Scientific).

Cloning of arbitrary PCR inserts into the plasmid backbone was performed by IDA as described previously (15). Library inserts were incubated in IDA reaction mixtures with 100 ng of plasmid backbone, and these reaction mixtures were transformed into electrocompetent *E. coli* ElectroMAX DH10B cells (ThermoFisher) and plated on medium with selective antibiotics. DNA from individual colonies was first screened by restriction digestion and sequenced to confirm that inserts of the desired size were incorporated. For generation of libraries for sorting, >50,000 colonies were collected from plates and mixed in LB selective medium and the culture was stored at –80°C. DNA extracted from this library was transformed into electrocompetent *E. coli* S17-1 λ pir cells containing a plasmid expressing *luxR* (pKM699). After this second transformation step, >50,000 colonies were collected from plates and mixed in LB selective medium and the culture was stored at –80°C.

Flow cytometry sorting. A BD FACSaria II was used to sort *E. coli* cells on the basis of expression of the fluorescent reporter. In preparation for sorting, the library culture from the frozen stock was grown in 50 ml of selective medium at 30°C with shaking to an OD₆₀₀ of 0.5. DNA was extracted from this culture as the input DNA sample for sequencing. Control cultures were used to set the sorting gates. The positive controls for maximal expression were strains containing pKM699 (expressing *luxR*) and a plasmid construct that demonstrated high levels of expression for P_{luxC} (pJV369, P_{luxC}-*gfp* positive-control plasmid) or P_{betI} (pCH50, P_{betI}-*mCherry* positive-control plasmid). The negative controls were strains containing pKM699 and pJS1194 (*gfp* negative-control plasmid) or pCH76 (*mCherry* negative-control plasmid), which lack promoters in front of *gfp* or *mCherry*, respectively. The library culture was sorted into bins with >10,000 cells per bin. For P_{luxC} sorting, there were four bins with no, low, medium, or high GFP expression. For P_{betI} sorting, there were two bins with high or no *mCherry* expression. The sorted cell cultures were incubated in 5 ml of selective medium at 30°C with shaking at 275 rpm and grown to stationary phase. Cultures were stored at –80°C, and the DNA was extracted for sequencing with the GeneJet Miniprep kit (Thermo Scientific).

Illumina library preparation and sequencing. DNA extracted from all sorted samples and input controls was purified over a Performa DTR gel filtration cartridge (Edge Biosystems). Next, the DNA was sheared with a Covaris S220 in 6- by 16-mm microtubes to an average size of 400 bp and analyzed on an Agilent 2200 TapeStation with D1000 ScreenTape. The sheared DNA samples (1 µg) were each treated with terminal deoxynucleotidyl transferase (TdT) in a tailing reaction mixture using TdT (Promega) and a mixture of dCTP and ddCTP (475 and 25 µM final concentrations, respectively) to generate a poly(C) tail. The reaction mixtures were incubated at 37°C for 1 h, heat inactivated at 75°C for 20 min, and cleaned over a Performa DTR gel filtration cartridge. Next, two rounds of PCR were performed to attach sequences for Illumina sequencing (Table S1). In round 1, a C tail-specific primer (olj376) and a gene-specific primer (CH060 for the *gfp* library and CH063 for the *mCherry* library) were used to amplify the promoter fragments that were cloned into the plasmid by the RAIL method. This PCR used *Taq* polymerase (NEB) under the following cycling conditions: an initial denaturation at 95°C for 2 min; 24 cycles of 95°C for 30 s, 58°C for 30 s, and 72°C for 2 min; and a final extension step of 72°C for 2 min. The round 2 PCR used a nested gene-specific primer (CH061 for the *gfp* library and CH064 for the *mCherry* library) to provide added specificity and also to append the linker sequence needed for Illumina sequencing. The second primer in the reaction mixture contained different barcodes for each sample to enable the libraries to be pooled and sequenced simultaneously (Table S1; BC37-44). The round 2 PCRs were performed with *Taq* polymerase and cycling as follows: an initial denaturation at 95°C for 2 min; 12 cycles of 95°C for 30 s, 52°C for 30 s, and 72°C for 2 min; and a final extension step of 72°C for 2 min. These final PCR products were examined by DNA gel electrophoresis, at which point a smear of products was visible on the gel.

Sequencing was performed on a NextSeq 500 with a NextSeq 75 reagent kit, 42- by 42-bp paired-end run parameters, and gene-specific primers (CH062 for the *gfp* library and CH065 for the *mCherry* library). Reads were checked with FastQC (v0.11.5; available at <http://www.bioinformatics.babraham.ac.uk/projects/fastqc>) to ensure that the quality of the data was high and that there were no noteworthy artifacts associated with the reads. The R1 read from paired-end Illumina NextSeq reads were quality and adapter trimmed with Trimmomatic (v0.33) (26) by using the following parameters: ILLUMINACLIP: <adapters>:3:20:6 LEADING:3 TRAILING:3 SLIDINGWINDOW:4:20 MINLEN:25. Reads were high quality, with 80 to 85% of them surviving trimming. Reads were mapped against the *V. campbellii* ATCC BAA-1116 genome (including molecules NC_009777, NC_009783, and NC_009784) with Bowtie 2.3.2 (27) and visualized with JBrowse (version 1.10.12) to analyze the alignment with the *betI* and *luxC* gene regions. Analysis of the promoter-seq data hinged on contrasting high-, medium-, low-, and no-expression cells with the null distribution. The null distribution was determined by sequencing cells collected without any sorting applied. The reads associated with the null and high-, medium-, low-, and no-expression data sets for the *luxC* library (or null and high- and no-expression data sets for the *betI* library) were normalized by transforming the data into the fraction of bases covered, which was defined as the depth of coverage at a particular base divided by the total depth of coverage over the particular promoter region (*betI*, bases 1361141 to 1362440 in NC_009784; *luxC*, bases 1424774 to 1426907 in NC_009784). Analysis was performed by plotting the log₂ ratio of the observed fraction of bases covered (for the high-, medium-, low-, or no-expression cell collection) over the expected (null) distribution.

SUPPLEMENTAL MATERIAL

Supplemental material for this article may be found at <https://doi.org/10.1128/JB.00724-17>.

SUPPLEMENTAL FILE 1, PDF file, 5.6 MB.

ACKNOWLEDGMENTS

We thank Ankur Dalia for assistance with the design of the Illumina sequencing protocol and helpful discussions. We also thank James Ford of the Indiana University Center for Genomics and Bioinformatics for Illumina sequencing experiments, bioinformatics analysis, and assistance with graphing of the results. We also acknowledge Christiane Hassel of the Indiana University Flow Cytometry Core Facility for assistance with FACSaria II analysis and sorting. We thank Ryan Fritts for assistance with statistical

analyses. We appreciate the use of Jason Tennesen's BioTek Cytation plate reader, and we thank Ryan Chaparian and Blake Petersen for assistance with experiments.

This work was supported by National Institutes of Health (NIH) grant 1R35GM124698-01 to J.C.V.K. and startup funds from Indiana University to J.C.V.K. and M.L.B.

REFERENCES

- Rosochacki SJ, Matejczyk M. 2002. Green fluorescent protein as a molecular marker in microbiology. *Acta Microbiol Pol* 51:205–216.
- Jiang T, Xing B, Rao J. 2008. Recent developments of biological reporter technology for detecting gene expression. *Biotechnol Genet Eng Rev* 25:41–75. <https://doi.org/10.5661/bger-25-41>.
- Andersen JB, Sternberg C, Poulsen LK, Bjorn SP, Givskov M, Molin S. 1998. New unstable variants of green fluorescent protein for studies of transient gene expression in bacteria. *Appl Environ Microbiol* 64:2240–2246.
- Arnone MI, Dmochowski IJ, Gache C. 2004. Using reporter genes to study cis-regulatory elements. *Methods Cell Biol* 74:621–652. [https://doi.org/10.1016/S0091-679X\(04\)74025-X](https://doi.org/10.1016/S0091-679X(04)74025-X).
- van Kessel JC, Ulrich LE, Zhulin IB, Bassler BL. 2013. Analysis of activator and repressor functions reveals the requirements for transcriptional control by LuxR, the master regulator of quorum sensing in *Vibrio harveyi*. *mBio* 4:e00278-13. <https://doi.org/10.1128/mBio.00378-13>.
- Chaparian RR, Olney SG, Hustmyer CM, Rowe-Magnus DA, van Kessel JC. 2016. Integration host factor and LuxR synergistically bind DNA to coactivate quorum-sensing genes in *Vibrio harveyi*. *Mol Microbiol* 101:823–840. <https://doi.org/10.1111/mmi.13425>.
- Chatterjee J, Miyamoto CM, Zouzoulas A, Lang BF, Skouris N, Meighen EA. 2002. MetR and CRP bind to the *Vibrio harveyi* lux promoters and regulate luminescence. *Mol Microbiol* 46:101–111. <https://doi.org/10.1046/j.1365-2958.2002.03128.x>.
- Meighen EA. 1991. Molecular biology of bacterial bioluminescence. *Microbiol Rev* 55:123–142.
- Miyamoto CM, Smith EE, Swartzman E, Cao JG, Graham AF, Meighen EA. 1994. Proximal and distal sites bind LuxR independently and activate expression of the *Vibrio harveyi* lux operon. *Mol Microbiol* 14:255–262. <https://doi.org/10.1111/j.1365-2958.1994.tb01286.x>.
- Swartzman E, Silverman M, Meighen EA. 1992. The luxR gene product of *Vibrio harveyi* is a transcriptional activator of the lux promoter. *J Bacteriol* 174:7490–7493. <https://doi.org/10.1128/jb.174.22.7490-7493.1992>.
- Reznikoff WS. 1992. The lactose operon-controlling elements: a complex paradigm. *Mol Microbiol* 6:2419–2422. <https://doi.org/10.1111/j.1365-2958.1992.tb01416.x>.
- Dunn TM, Hahn S, Ogden S, Schleif RF. 1984. An operator at –280 base pairs that is required for repression of araBAD operon promoter: addition of DNA helical turns between the operator and promoter cyclically hinders repression. *Proc Natl Acad Sci U S A* 81:5017–5020.
- Galagan J, Lyubetskaya A, Gomes A. 2013. ChIP-Seq and the complexity of bacterial transcriptional regulation. *Curr Top Microbiol Immunol* 363:43–68.
- van Kessel JC, Rutherford ST, Cong JP, Quinodoz S, Healy J, Bassler BL. 2015. Quorum sensing regulates the osmotic stress response in *Vibrio harveyi*. *J Bacteriol* 197:73–80. <https://doi.org/10.1128/JB.02246-14>.
- Gibson DG, Young L, Chuang RY, Venter JC, Hutchison CA, III, Smith HO. 2009. Enzymatic assembly of DNA molecules up to several hundred kilobases. *Nat Methods* 6:343–345. <https://doi.org/10.1038/nmeth.1318>.
- Welsh J, McClelland M. 1990. Fingerprinting genomes using PCR with arbitrary primers. *Nucleic Acids Res* 18:7213–7218. <https://doi.org/10.1093/nar/18.24.7213>.
- Williams JG, Kubelik AR, Livak KJ, Rafalski JA, Tingey SV. 1990. DNA polymorphisms amplified by arbitrary primers are useful as genetic markers. *Nucleic Acids Res* 18:6531–6535. <https://doi.org/10.1093/nar/18.22.6531>.
- Showalter RE, Martin MO, Silverman MR. 1990. Cloning and nucleotide sequence of luxR, a regulatory gene controlling bioluminescence in *Vibrio harveyi*. *J Bacteriol* 172:2946–2954. <https://doi.org/10.1128/jb.172.6.2946-2954.1990>.
- van Kessel JC, Rutherford ST, Shao Y, Utria AF, Bassler BL. 2013. Individual and combined roles of the master regulators AphA and LuxR in control of the *Vibrio harveyi* quorum-sensing regulon. *J Bacteriol* 195:436–443. <https://doi.org/10.1128/JB.01998-12>.
- Park JS, Marr MT, Roberts JW. 2002. E. coli Transcription repair coupling factor (Mfd protein) rescues arrested complexes by promoting forward translocation. *Cell* 109:757–767. [https://doi.org/10.1016/S0092-8674\(02\)00769-9](https://doi.org/10.1016/S0092-8674(02)00769-9).
- Epshtein V, Toulme F, Rahmouni AR, Borukhov S, Nudler E. 2003. Transcription through the roadblocks: the role of RNA polymerase cooperation. *EMBO J* 22:4719–4727. <https://doi.org/10.1093/emboj/cdg452>.
- Waters CM, Bassler BL. 2006. The *Vibrio harveyi* quorum-sensing system uses shared regulatory components to discriminate between multiple autoinducers. *Genes Dev* 20:2754–2767. <https://doi.org/10.1101/gad.1466506>.
- Teng SW, Wang Y, Tu KC, Long T, Mehta P, Wingreen NS, Bassler BL, Ong NP. 2010. Measurement of the copy number of the master quorum-sensing regulator of a bacterial cell. *Biophys J* 98:2024–2031. <https://doi.org/10.1016/j.bpj.2010.01.031>.
- Lin B, Wang Z, Malanoski AP, O'Grady EA, Wimpee CF, Vuddhakul V, Alves N, Jr, Thompson FL, Gomez-Gil B, Vora GJ. 2010. Comparative genomic analyses identify the *Vibrio harveyi* genome sequenced strains BAA-1116 and HY01 as *Vibrio campbellii*. *Environ Microbiol Rep* 2:81–89. <https://doi.org/10.1111/j.1758-2229.2009.00100.x>.
- Schmidt KH, Pennaneach V, Putnam CD, Kolodner RD. 2006. Analysis of gross-chromosomal rearrangements in *Saccharomyces cerevisiae*. *Methods Enzymol* 409:462–476. [https://doi.org/10.1016/S0076-6879\(05\)09027-0](https://doi.org/10.1016/S0076-6879(05)09027-0).
- Bolger AM, Lohse M, Usadel B. 2014. Trimmomatic: a flexible trimmer for Illumina sequence data. *Bioinformatics* 30:2114–2120. <https://doi.org/10.1093/bioinformatics/btu170>.
- Langmead B, Salzberg SL. 2012. Fast gapped-read alignment with Bowtie 2. *Nat Methods* 9:357–359. <https://doi.org/10.1038/nmeth.1923>.